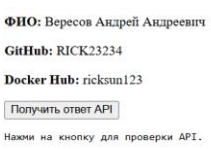
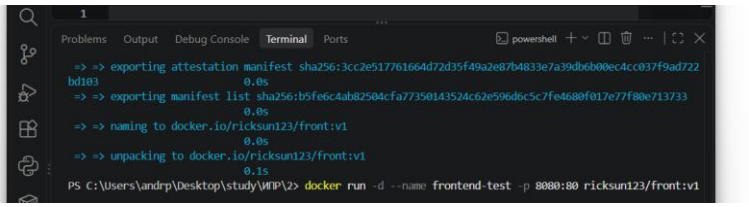
1.1

```
PS C:\Users\andrp\Desktop\study\ИПР\2> docker build -t ricksun123/front:v1 .
[+] Building 32.4s (10/10) FINISHED                                                                                 docker:des
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 202B
 => [internal] load metadata for docker.io/library/nginx:alpine
 => [auth] library/nginx:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/4] FROM docker.io/library/nginx:alpine@sha256:892bb2152464f0c5d8972593a5f5afaf8054552580b699d95f8ff4af08c2d33d
 => => resolve docker.io/library/nginx:alpine@sha256:892bb2152464f0c5d8972593a5f5afaf8054552580b699d95f8ff4af08c2d33d
 => => sha256:1fdd0614a47ec2896aea36fb280a7b5e8655614dfb56ec1a64002131791de601 20.53MB / 20.53MB
 => => sha256:6638b436592c96dd194db76a6b786b60b344d80ebe160e646c871d4e7eff25b5 1.40kB / 1.40kB
 => => sha256:70683d58fa5c20b5b31c9998a16f6634720d3393dc62934a608754abe449fe3b 1.21kB / 1.21kB
 => => sha256:710ec83ca6cd71fa5636d861343b5660636728f71992fd1b68914e03c7af3ad4 405B / 405B
 => => sha256:b44956913298a55652fa97af34d296339c089250911e46e1a1c0eab8a88538c7 966B / 966B
 => => sha256:3bf3671a4e6a18510ca49e3a4a226c2ef734a646e01e34e1ad0eff5faa01e3a0 629B / 629B
 => => sha256:6711e0c4f5a79e8db637a3412394f29da647e54476d67c5b211a11527568ddc5 4.42MB / 4.42MB
 => => sha256:e2de96513ba9eb53b431787ec8a65cdde380ac4772a3e4c4b714dcfde2a102b5 3.85MB / 3.85MB
 => => extracting sha256:e2de96513ba9eb53b431787ec8a65cdde380ac4772a3e4c4b714dcfde2a102b5
 => => extracting sha256:6711e0c4f5a79e8db637a3412394f29da647e54476d67c5b211a11527568ddc5
 => => extracting sha256:3bf3671a4e6a18510ca49e3a4a226c2ef734a646e01e34e1ad0eff5faa01e3a0
 => => extracting sha256:b44956913298a55652fa97af34d296339c089250911e46e1a1c0eab8a88538c7
 => => extracting sha256:710ec83ca6cd71fa5636d861343b5660636728f71992fd1b68914e03c7af3ad4
 => => extracting sha256:70683d58fa5c20b5b31c9998a16f6634720d3393dc62934a608754abe449fe3b
 => => extracting sha256:6638b436592c96dd194db76a6b786b60b344d80ebe160e646c871d4e7eff25b5
 => => extracting sha256:1fdd0614a47ec2896aea36fb280a7b5e8655614dfb56ec1a64002131791de601
 => [internal] load build context
 => => transferring context: 1.11kB
 => [2/4] RUN apk add --no-cache curl
 => [3/4] COPY index.html /usr/share/nginx/html/index.html
 => [4/4] COPY nginx.conf /etc/nginx/conf.d/default.conf
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:260f154ad43418bfd0468985ed25b112aa2f23be05780769147f045d7265adda
 => => exporting config sha256:3734adefe6050922d00cf10944904951c151f2adeafda4a239c4b3d1cdd13381
 => => exporting attestation manifest sha256:3cc2e517761664d72d35f49a2e87b4833e7a39db6b00ec4cc037f9ad722bd103
 => => exporting manifest list sha256:b5fe6c4ab82504cfa77350143524c62e596d6c5c7fe4680f017e77f80e713733
 => => naming to docker.io/ricksun123/front:v1
 => => unpacking to docker.io/ricksun123/front:v1
```

1.2

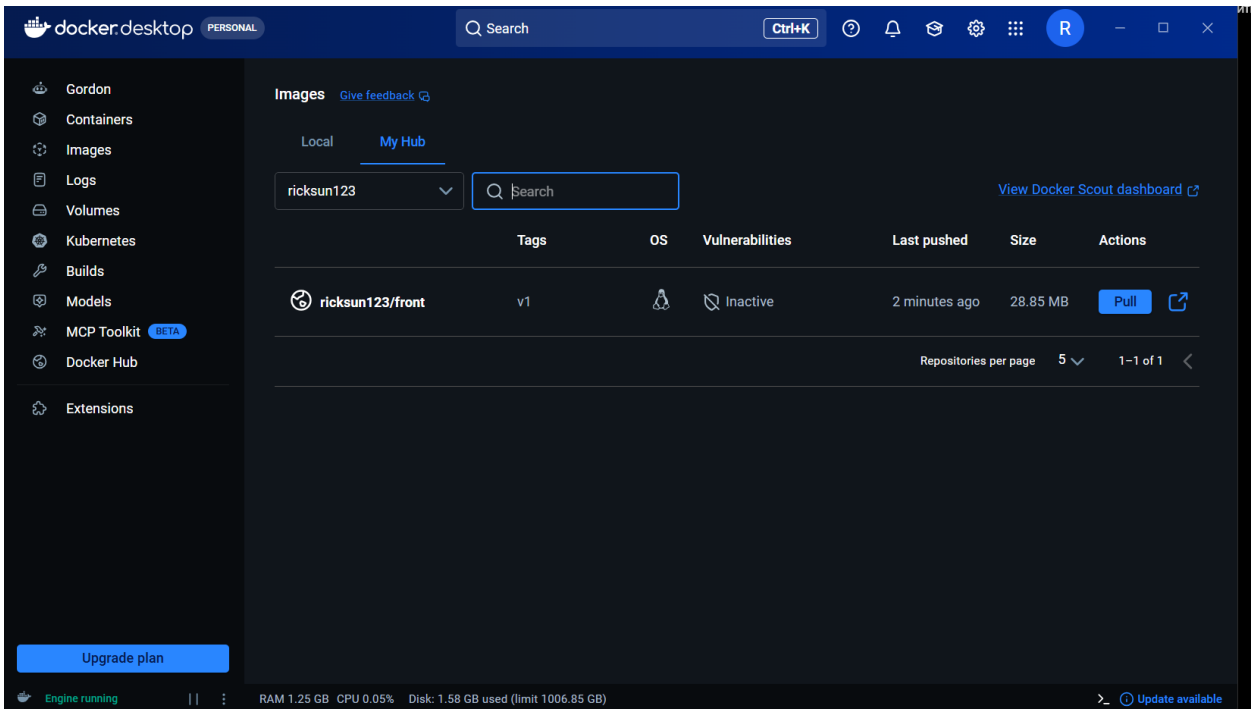




1.3

```
PS C:\Users\andrp\Desktop\study\ИПР\2> docker push ricksun123/front:v1
The push refers to repository [docker.io/ricksun123/front]
44136fa355b3: Pushed
6711e0c4f5a7: Pushed
e2de96513ba9: Pushed
15576ac1d4db: Pushed
3bf3671a4e6a: Pushed
710ec83ca6cd: Pushed
b44956913298: Pushed
70683d58fa5c: Pushed
6638b436592c: Pushed
1fdd0614a47e: Pushed
4b8f9aa3cecb: Pushed
8b68eefd1c53: Pushed
b0652ebc41db: Pushed
v1: digest: sha256:b5fe6c4ab82504cfa77350143524c62e596d6c5c7fe4680f017e77f80e713733 size: 856
```

1.4



1.5

```
PS C:\Users\andrp\Desktop\study\ИПР\2> docker rmi ricksun123/front:v1
Untagged: ricksun123/front:v1
Deleted: sha256:b5fe6c4ab82504cfa77350143524c62e596d6c5c7fe4680f017e77f80e713733
PS C:\Users\andrp\Desktop\study\ИПР\2> docker pull ricksun123/front:v1
v1: Pulling from ricksun123/front
44136fa355b3: Download complete
b0652ebc41db: Pull complete
4b8f9aa3cecb: Pull complete
3b68eefd1c53: Pull complete
15576ac1d4db: Download complete
Digest: sha256:b5fe6c4ab82504cfa77350143524c62e596d6c5c7fe4680f017e77f80e713733
Status: Downloaded newer image for ricksun123/front:v1
docker.io/ricksun123/front:v1
```

2.1

```yaml
docker-compose.yml
Run All Services
services:
  Run Service
  frontend:
    build: .
    image: ricksun123/front:v1
    ports:
      - "8080:80"
    depends_on:
      - api

  Run Service
  api:
    image: containous/whoami
    expose:
      - "80"

  Run Service
  logger:
    image: busybox
    command: sh -c "while true; do date '+%Y-%m-%d %H:%M:%S' >> /data/log.txt; sleep 5; done"
    volumes:
      - logger-data:/data

volumes:
  logger-data:
```

2.2

```
PS C:\Users\andrp\Desktop\study\ИПР\2> docker compose up -d --build
[+] up 7/7
 ✓ Image busybox           Pulled                                                    9.8s
 ✓ Image containous/whoami Pulled                                                    5.4s
[+] Building 4.3s (12/12) FINISHED
 => [internal] load local bake definitions                                          0.0s
 => => reading from stdin 544B                                                      0.0s
 => [internal] load build definition from Dockerfile                                0.0s
 => => transferring dockerfile: 202B                                                0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                     3.0s
 => [auth] library/nginx:pull token for registry-1.docker.io                        0.0s
 => [internal] load .dockerignore                                                   0.0s
 => => transferring context: 2B                                                     0.0s
 => [1/4] FROM docker.io/library/nginx:alpine@sha256:892bb2152464f0c5d8972593a5f5afaf8054552580b6  0.0s
 => => resolve docker.io/library/nginx:alpine@sha256:892bb2152464f0c5d8972593a5f5afaf8054552580b6  0.0s
 => [internal] load build context                                                   0.0s
 => => transferring context: 62B                                                    0.0s
 => CACHED [2/4] RUN apk add --no-cache curl                                        0.0s
 => CACHED [3/4] COPY index.html /usr/share/nginx/html/index.html                   0.0s
 => CACHED [4/4] COPY nginx.conf /etc/nginx/conf.d/default.conf                     0.0s
 => exporting to image                                                              0.2s
 => => exporting layers                                                             0.0s
 => => exporting manifest sha256:57ee94cd10259aeddfe4f6c4f7bc4e6ed8ff93a55260aac94d0e8d6a798f3387  0.0s
 => => exporting config sha256:810eea3dc9ab032a50788187897ce42407e219f535c974bd9ed7f6f94cd0af13  0.0s
 => => exporting attestation manifest sha256:ff97ce4fe5f59924b49a351c474752cfc62da75ab42c136f4270  0.0s
 => => exporting manifest list sha256:08a18baea427a374219ead6d1a4ea477029704c984c272ed02471efed61  0.0s
[+] up 13/13g to docker.io/ricksun123/front:v1                                      0.0s
 ✓ Image busybox           Pulled                                                    9.8s
 ✓ Image containous/whoami Pulled                                                    5.4s
 ... 6 more
```

```
PS C:\Users\andrp\Desktop\study\ИПР\2> docker compose ps
NAME           IMAGE                  COMMAND                  SERVICE    CREATED         STATUS         PORTS
2-api-1        containous/whoami      "/whoami"                api        5 seconds ago   Up 5 seconds   80/tcp
2-frontend-1   ricksun123/front:v1    "/docker-entrypoint.…"   frontend   5 seconds ago   Up 4 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
2-logger-1     busybox                "sh -c 'while true; …"   logger     5 seconds ago   Up 5 seconds
```

2.3

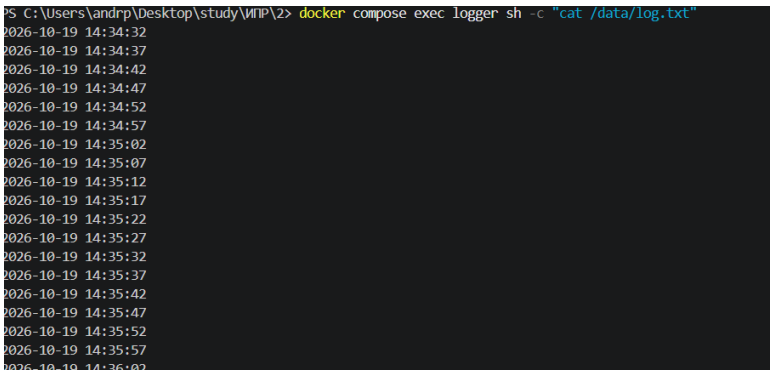
```
PS C:\Users\andrp\Desktop\study\ИПР\2> docker compose exec frontend sh
/ # curl http://api
Hostname: 0cc7d9890d4d
IP: 127.0.0.1
IP: ::1
IP: 172.18.0.3
RemoteAddr: 172.18.0.4:46862
GET / HTTP/1.1
Host: api
User-Agent: curl/8.22.0
Accept: */*
```

2.4

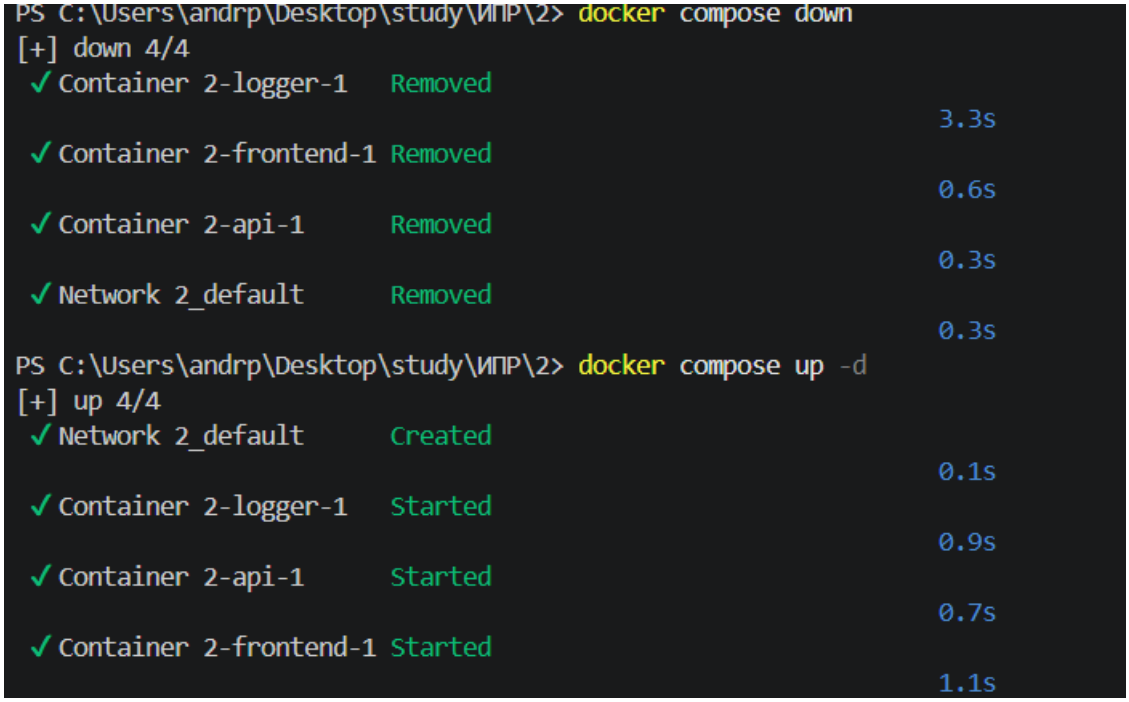
```
PS C:\Users\andrp\Desktop\study\ИПР\2> docker volume ls | Select-String logger

local     2_logger-data
```

2.5


PS C:\Users\andrp\Desktop\study\ИПР\2> docker compose exec logger sh -c "cat /data/log.txt"
2026-10-19 14:34:32
2026-10-19 14:34:37
2026-10-19 14:34:42
2026-10-19 14:34:47
2026-10-19 14:34:52
2026-10-19 14:34:57
2026-10-19 14:35:02
2026-10-19 14:35:07
2026-10-19 14:35:12
2026-10-19 14:35:17
2026-10-19 14:35:22
2026-10-19 14:35:27
2026-10-19 14:35:32
2026-10-19 14:35:37
2026-10-19 14:35:42
2026-10-19 14:35:47
2026-10-19 14:35:52
2026-10-19 14:35:57

3.1


PS C:\Users\andrp\Desktop\study\ИПР\2> docker compose down
[+] down 4/4
✓ Container 2-logger-1 Removed
3.3s
✓ Container 2-frontend-1 Removed
0.6s
✓ Container 2-api-1 Removed
0.3s
✓ Network 2_default Removed
0.3s
PS C:\Users\andrp\Desktop\study\ИПР\2> docker compose up -d
[+] up 4/4
✓ Network 2_default Created
0.1s
✓ Container 2-logger-1 Started
0.9s
✓ Container 2-api-1 Started
0.7s
✓ Container 2-frontend-1 Started
1.1s

3.2

```
PS C:\Users\andrp\Desktop\study\ИПР\2> docker compose exec logger sh -c "cat /data/log.txt"
2026-10-19 14:34:32
2026-10-19 14:34:37
2026-10-19 14:34:42
2026-10-19 14:34:47
2026-10-19 14:34:52
2026-10-19 14:34:57
2026-10-19 14:35:02
2026-10-19 14:35:07
2026-10-19 14:35:12
2026-10-19 14:35:17
2026-10-19 14:35:22
2026-10-19 14:35:27
2026-10-19 14:35:32
2026-10-19 14:35:37
2026-10-19 14:35:42
2026-10-19 14:35:47
2026-10-19 14:35:52
PS C:\Users\andrp\Desktop\study\ИПР\2> docker compose down -v
[+] down 5/5
 ✓ Container 2-frontend-1 Removed
 ✓ Container 2-logger-1   Removed
 ✓ Container 2-api-1      Removed
 ✓ Volume 2_logger-data   Removed
 ✓ Network 2_default      Removed
PS C:\Users\andrp\Desktop\study\ИПР\2> docker compose up -d
[+] up 5/5
 ✓ Network 2_default      Created
 ✓ Volume 2_logger-data   Created
 ✓ Container 2-logger-1   Started
 ✓ Container 2-api-1      Started
 ✓ Container 2-frontend-1 Started
PS C:\Users\andrp\Desktop\study\ИПР\2> docker compose exec logger sh -c "cat /data/log.txt"
2026-10-19 14:45:31
2026-10-19 14:45:36
2026-10-19 14:45:41
```